Snowflake CoCo (Cortex Code) — Cheat Sheet

Active reference · compiled 2026-10-09 · verify with /changelog & docs

1 · THREE SURFACES

	Snowsight	Desktop	CLI
Form	Panel (CoCo icon, lower-right)	AI IDE (mac/Win)	`cortex` terminal agent
Install	None	Download installer	`curl ... install.sh`
Best for	Analysts: SQL, discovery, admin Q&A, notebooks	Parallel agents, charts, apps, dbt, SQL Playground, notebooks	Scripts, CI (`exec`), ACP, `--cloud`, automations
Files/Git	Workspace only	Local	Local (+cloud sandbox)
Daily-limit param	`CORTEX_CODE_SNOWSIGHT_...`	`CORTEX_CODE_DESKTOP_...`	`CORTEX_CODE_CLI_...`

All three listed GA. Param suffix: `DAILY_EST_CREDIT_LIMIT_PER_USER`. Desktop/CLI: commercial non-Gov/VPS/Sovereign accounts; Snowsight also FedRAMP/DoD/KSA. **Cross-region inference must be enabled.**

2 · ACCESS CHECKLIST

Need	Grant / setting
CLI / Desktop	`SNOWFLAKE.CORTEX_USER` (default to PUBLIC) **or** `CORTEX_AGENT_USER` (least privilege)
Snowsight	`SNOWFLAKE.COPILOT_USER` **+** one of the above
Blocked?	Daily limit `0` blocks. Access = role **AND** non-zero limit
Automations	`EXECUTE AGENT TASK` (default PUBLIC)
Audit views	`AI_OBSERVABILITY_READER`; `IMPORTED PRIVILEGES ON DATABASE SNOWFLAKE`
Models	`SHOW CORTEX BASE MODELS IN SCHEMA SNOWFLAKE.MODELS`

```
ALTER ACCOUNT SET CORTEX_ENABLED_CROSS_REGION = 'AWS_US';
-- or 'ANY_REGION'  (ACCOUNTADMIN; check residency policy)
```

Trial accounts: AI off until admin adds a credit card. Standard trial accts unsupported for CLI. Opted out of legacy Copilot => Snowsight CoCo disabled.

3 · INSTALL & CONNECT

CLI

```
curl -LsS https://ai.snowflake.com/static/cc-scripts/install.sh | sh
# if ~/.curlrc breaks it: curl -qLsS ...
irm https://ai.snowflake.com/static/cc-scripts/install.ps1 | iex   # Windows
cortex --version     cortex update     cortex -c MYCONN
```

```
# ~/.snowflake/connections.toml   (chmod 600)
default_connection_name = "dev"
[dev]
account = "myorg-myaccount"
user = "me"
authenticator = "externalbrowser"   # or oauth_authorization_code
role = "MY_ROLE"
warehouse = "COMPUTE_WH"
```

Key-pair: `authenticator = "snowflake_jwt"` + `private_key_path`. Same file as Snowflake CLI (`snow`). Platforms: macOS arm64/x64, Linux x64/arm64, Windows native x64 or WSL.

Desktop

- First run: Welcome > Connect (auto-detects `connections.toml`) > Agent/Editor view > theme.
- Auth: **Local OAuth (recommended)**, SSO external browser, password, key-pair via TOML. Tokens in OS keychain.
- Top bar connection menu: Change Role, Default Warehouse, **Private Mode**. Win installer: User (no admin) vs System.
- Logs (mac): `~/Library/Application Support/Cortex Code/logs/`. Cmd Palette: **Reset Onboarding**.

Snowsight

- Starts on your **default role** - fix with `ALTER USER ... SET DEFAULT_ROLE` or ask CoCo to switch roles.
- `@` attach tables/schemas/views; `/` skills; highlight SQL: Quick Edit / Format / Add to Chat / Explain; **Fix** button on errors; Tab accepts inline suggestions.

4 · MODES & APPROVALS

Surface	Modes	Switch
CLI	**Confirm** (default) · **Plan** · **Bypass**	`Shift-Tab`, `/plan` `/plan-off` `/bypass`, `Ctrl+P`
Desktop	Default Approvals · Bypass (machine-wide). Chat: Ask / Edit / Agent (+Plan)	Picker bottom-left; `Shift+Tab` Agent<>Plan
ACP	`standard` `plan` `bypass`	session option

- **Plan mode** = read-only research (read files, `SELECT/SHOW/DESCRIBE`, web) -> plan card -> **Build** (Desktop Cmd+Shift+B). Plans saved to `.snowflake/cortex/plans/`.
- CLI risk tiers: SAFE auto > LOW > MEDIUM prompts > HIGH (`rm` `curl` `wget` `sudo`) always prompts > CRITICAL (`rm -rf`). SQL: READ_ONLY auto; WRITE & USE_ROLE prompt.
- Desktop prompts: Allow once / session / Always / Reject. `CALL` may re-prompt. Reset: **Reset Tool Confirmations**.

Never bypass on prod. Prod habit: `/plan` first, read-only role, RSS (below).

- `COCO_DANGEROUS_MODE_REQUIRE_SQL_WRITE_PERMISSION=true` keeps SQL-write prompts even in Bypass.

5 · RESTRICTED SESSION SCOPE (RSS)

```
/guardrails                # restrict this session / create scope
/guardrails status
cortex --with-restricted-session-scope=READONLY_PM
cortex --sql-read-only       # /sql-writes toggles
```

- Options: read-only or role allow/block list; stored in `USER$<user>.RSS`.
- **Once on, you can switch RSS but never remove/relax it - start a new session.** Not supported with `--cloud`.
- Desktop: **+ > Restrict this session**. Snowsight: same via chat.

6 · MODELS & CONTEXT

- **Auto** (Snowflake picks) · **Auto Intelligent** (best, costliest) · **Auto Efficient** (cheapest). `/model` or `--model auto`.
- Effort: CLI `--effort minimal|low|medium|high|max`; Snowsight Low/Med/High.
- Org default model: agent `CORTEX_CODE.CONFIG.DEFAULT` with `models: { orchestration: auto }`; GRANT USAGE to roles.
- Instruction files: `AGENTS.md`, `CLAUDE.md`, `CORTEX.md`, `SNOWFLAKE.md`, `RULES.md`, `.cursorrules`, `.claude/rules/*.md` (cap ~100k chars; workspace > user > profile).
- `/compact` frees context. Desktop memory: `~/.snowflake/cortex/memory/` (`MEMORY.md`).

7 · FILE MAP

```
~/.snowflake/connections.toml
~/.snowflake/cortex/
  settings.json  mcp.json  hooks.json  permissions.json
  conversations/ skills/ agents/ plugins/ profiles/ logs/
project: .cortex/ (skills agents plugins settings.json mcp.json)
         .claude/ is read too (Claude-compat)
```

`cortex --private` = no history saved. `/clear` starts a NEW conversation (1.1.65+); `/cls` or Ctrl+L clears screen only.

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8 · CLI FLAGS

Flag	Purpose
`-c conn` · `-w dir` · `-m model`	Connection / workdir / model
`--continue` · `-r [id|last]`	Resume latest / by id (picker if blank)
`-p "prompt"` · `--output-format stream-json`	Batch print & exit; scriptable
`--plan` · `--auto-accept-plans`	Require approval / auto-accept plans
`--bypass`	Auto-approve ALL (risky)
`--allowed-tools "Bash(git *)"`	Pre-approve tools
`--disallowed-tools`	Hide tools from model
`--max-turns n` · `--effort`	Limit turns / thinking
`--private`	No history / no server logging
`--cloud [ws]` `--no-workspace` `--github SECRET`	Run tools in Snowflake container
`--plugin-dir` · `--only-explicit-{mcp-servers,plugins,skills}`	Control what loads
`--mcp-wait` · `--fork` · `--mode code`	Wait for MCP / open shared convo / lean toolset

Subcommands

```
cortex exec "prompt" | --file f | --file -     # CI/CD, non-interactive
cortex mcp list|add|get|remove|reconnect|start
cortex skill list|add|update|publish|remove|sync
cortex plugin install|uninstall|enable|disable|list|validate|update
cortex automation create|list|describe|doctor|execute|suspend|resume|drop
cortex conversations list|search|delete|transcript
cortex acp serve -c CONN       cortex auth login|status
cortex managed-settings        cortex profile publish | cortex ws
```

Exit codes: 0 ok, non-zero fail (details on stderr). `exec`: plan mode off, prompts auto-rejected.

9 · SLASH COMMANDS (interactive)

Group	Commands
Session	`/help` `/new` `/rename` `/resume` `/rewind` `/fork` `/clear` `/quit` `/restart`
Mode/model	`/model` `/plan` `/plan-off` `/bypass` `/bypass-off` `/status` `/stats`
Data	`/sql q [--limit n]` `/table` `/connections` `/guardrails` `/lineage` `/fdbt`
Dev	`/sh` or `!cmd` `/diff` `/worktree create|list|switch|delete` `/add-dir`
Config	`/settings` `/theme` `/sandbox` `/workspace` (cloud) `/index` `/compact`
Extend	`/skill` `/mcp` `/hooks` `/commands` `/agents` `/plugin` `/automation` `/monitors`
Share	`/share-skill-and-plugin` `/find-skill-and-plugin` `/import-claude-config`
Misc	`/feedback` (local .tgz) `/changelog` `/update` `/wipe-session` `/docs`

10 · SHORTCUTS

Key	Action	Key	Action
`Ctrl+C`	Cancel (x2 exit)	`Ctrl+J`	Newline
`Ctrl+L`	Clear screen	`Ctrl+T`	Table viewer
`Shift-Tab`	Cycle modes	`Ctrl+B`	Monitor agents
`Ctrl+Q`	Prompt in $EDITOR	`Esc-u`	Undo /rewind
`Tab`	Complete	`Up/Down`	History

11 · SKILLS ($name in CLI, / in Desktop & Snowsight)

```
---
name: my-skill
description: when/why to use it (drives auto-invoke)
tools:
- sql_execute        # CLI name; Desktop = snowflake_sql_execute
- bash               # tool names LOWERCASE, case-sensitive
---
# Instructions ... Examples: User: $my-skill ...
```

- Precedence: project `.cortex/skills` or `.claude/skills` > user `~/.snowflake/cortex/skills` > global > session > remote > bundled.
- Share: **Catalog** `snow://skill_catalog/DB.SCH.EXT[/versions/version$1]`, Git URL, or stage:

```
cortex skill publish ./skills --to-stage @DB.SCH.STG/skills/
cortex skill add @DB.SCH.STG/skills/    # stage needs READ (add), WRITE (publish)
```

- Bundled seen: `$dynamic-tables` `$semantic-view` `$cortex-agent` `$agent-optimize` `$skill-development` `plugin-creator` `find-skill` `share-skill`. Catalog = same account only, **no security scan**.

12 · SUBAGENTS & HOOKS

- Built-in: `general-purpose` `explore` `plan` `feedback` `data-discovery`. Custom: `.cortex/agents/x.md` (frontmatter `name` `description` `tools` `model: auto|inherit|id`).

Tool list must be lowercase (`bash`, not `Bash`) - wrong case = agent with NO tools, silently.

- Max 50 concurrent background agents; `/agents` or Ctrl+B; worktree isolation -> branch `agent/<id>`.

```
{"hooks":{"PreToolUse":[{"matcher":"sql_execute",
  "hooks":[{"type":"command","command":"bash .cortex/hooks/chk.sh","timeout":60}]}]}}
```

- Events: PreToolUse*, PostToolUse, PermissionRequest*, UserPromptSubmit, SessionStart/End, PreCompact, Stop, SubagentStop, Notification, Setup (*can block). **Exit 2 = block**, 0 = allow. JSON on stdin/stdout. Priority: local > project > user > global.

13 · MCP

```
cortex mcp add git uvx mcp-server-git
cortex mcp add api https://api.example.com --type http -H "Authorization: Bearer ${TOKEN}"
cortex mcp start            # /mcp in session
```

- Transports: stdio, http (streamable), sse. Config `~/.snowflake/cortex/mcp.json` key `mcpServers`; `${VAR}` `${VAR:-default}`; OAuth default port **8585**; secrets move to OS keychain.
- Tool name `mcp__<server>__<tool>` (<=64 chars). Limits: result 50 KB, timeout 60 s (`COCO_MCP_TOOL_TIMEOUT_MS`).
- Precedence: admin-enforced > profile > user > plugin. `areUserMcpServersAllowed:false` kills user MCP.

14 · PLUGINS · ACP · SDK

- **Plugin** = `.cortex-plugin/plugin.json` (`name` required) + `skills/` `agents/` `commands/` `hooks/` `.mcp.json`. Dev loop: `cortex plugin validate ./p` then `cortex --plugin-dir ./p`. Not signed - review before install.
- **ACP** (Zed, JetBrains, Neovim): `cortex acp serve -c CONN`. JetBrains needs absolute path. stdout noise breaks it. VS Code -> Snowflake extension.

```
pip install cortex-code-agent-sdk      # Py >=3.10
npm install cortex-code-agent-sdk      # Node >=22   (Preview)
```

15 · AUTOMATIONS & CLOUD SANDBOX

```
cortex automation create --name pipe_health \
  --prompt "Summarize failed runs, last 24h" \
  --schedule "daily at 9am" --timezone America/New_York
cortex automation doctor pipe_health   # state, errors, query id
cortex automation execute pipe_health --wait
```

- Hosted automation = `AGENT TASK` in `USER$.PUBLIC`; runs as **you** with **default role + default secondary roles**; no warehouse; **no interactive approvals** -> least-privilege role.
- Limits (preview): min **hourly**, history/threads **2 months**, time-based only, no local files/MCP (use stages + `--mcp`), not in gov/FedRAMP/DoD/VPS/China.
- Desktop local automations: only while app open & machine awake; 50-run history; latest missed run caught up.

```
cortex --cloud                # default workspace
USER$<u>.PUBLIC.DEFAULT$
cortex --cloud DB.SCH.WS --github DB.SCH.PAT_SECRET
cortex exec --cloud "prompt" --allowed "Read"   # CI
```

- Only `/workspace` persists. Egress restricted; `ALTER ACCOUNT SET COCO_CLOUD_AGENTS_NON_SNOWFLAKE_EGRESS_DISABLED = TRUE;`. No RSS / `--sql-read-only` / local skills in cloud.

16 · SECURITY

Layer	Control
Identity	Runs as you; RBAC, masking, row policies apply
Org policy	`managed-settings.json` (mac `/Library/Application Support/Cortex/`, Linux `/etc/cortex/`, Win `%ProgramData%\Cortex\`) or MDM
Verify	`cortex managed-settings` (bad JSON => restricted mode)
Isolation	`/sandbox on`; Desktop `chat.agent.sandbox.enabled`; `--cloud`
Trust	Desktop Workspace Trust gate; plugins/skills/MCP unsigned

```
managed-settings: version "1.0";
permissions{onlyAllow,deny,defaultMode,
  dangerouslyAllowAll};
settings{forceNoHistoryMode,forceSandboxEnabled};
  required{minimumVersion}; files{connectionsFile,mcpFile}; ui{...}
eval: managed deny > profile deny > managed onlyAllow > profile
onlyAllow > default
```

- Best practice: `externalbrowser`/OAuth; PAT <=90 days; files 600 / dirs 700; role-per-env; **never ACCOUNTADMIN for routine work**; trusted MCP only.
- Treat untrusted READMEs/web/tool output as injection vectors (PromptArmor flaw fixed in CLI 1.0.25). Leaked PAT: revoke -> new token in env var -> review QUERY_HISTORY.

17 · COST & OBSERVABILITY

```
-- -1 unlimited (default) | 0 blocked | N = block when est. usage in
last 24h > N
ALTER ACCOUNT SET CORTEX_CODE_CLI_DAILY_EST_CREDIT_LIMIT_PER_USER =
20;
ALTER USER pwr SET CORTEX_CODE_CLI_DAILY_EST_CREDIT_LIMIT_PER_USER =
50;
ALTER USER pwr UNSET
CORTEX_CODE_CLI_DAILY_EST_CREDIT_LIMIT_PER_USER;
```

- Per-surface, rolling 24 h; user value overrides account. Error: "Daily credit usage limit reached." Also: **Budgets** (tag users) & **per-user quotas** (monthly/weekly/daily, auto-block).
- Billing: AI Credits ($2.00 global / $2.20 regional routing per pricing page); no per-seat fee; warehouse & storage separate. Token rates: Service Consumption Table.

```
SNOWFLAKE.LOCAL.AI_OBSERVABILITY_EVENTS             -- spans
(CodingAgentRun, CodingAgent.Step-0)
SNOWFLAKE.ACCOUNT_USAGE.SNOWFLAKE_COCO_USAGE_HISTORY  --
credits/tokens (INTERFACE, REQUEST_ID)
...CORTEX_CODE_{CLI|DESKTOP|SNOWSIGHT}_USAGE_HISTORY
```

Use the **usage views** (not the event table) as source of truth for credits; event delivery is best-effort.

18 · DESKTOP FEATURE POCKET GUIDE

Feature	Remember
Charts	line/area/bar/stacked/scatter/pie/metric cards; <=500 rows; Desktop-only; exploratory, not BI
SQL Playground	Cmd/Ctrl+Enter run; **Fix** returns diff (Keep/Undo); sheets in `~/.snowflake/cortex/sql/`
dbt	Local or Snowflake-managed (`EXECUTE DBT PROJECT FROM WORKSPACE`, needs personal DBs); lineage > Add to Chat
Apps	Streamlit (`snow streamlit deploy`) or App Runtime (Node 22+, `snow app deploy`); Caller's vs Owner's Rights
Notebook	`.ipynb` + kernel; Keep/Undo per cell
Browser	Cmd+E toggles Agent/Editor view; works on localhost
tgrep	Semantic code search; opt-in per workspace; not on remote/SSH

19 · ROLE QUICK-PROMPTS

Data Analyst

- "What databases/schemas can I read with claims data? Which have PII tags or masking?"
- "Write a query: monthly paid amount by line of business, 12 months. Show SQL + assumptions (grain, filters)."
- "Use @models/revenue.yaml to answer: what was revenue last month?"

System Analyst

- "Show lineage from RAW_DB.ORDERS to downstream dashboards."
- "Compare BRONZE_A vs BRONZE_B: equivalent fields, type fixes, open questions."
- "Plan (wait for approval): Dynamic Table SILVER_X from A and B; include validation queries + rollback note."

Integration Architect

- "List MCP servers/skills/plugins loaded, their origin and which win on name collision."
- "Review this hook/managed-settings against least privilege; list gaps."

Always ask for: **SQL used + assumptions + validation query**. Use fully qualified names (`DB.SCHEMA.TABLE`).

20 · TROUBLESHOOTING

Symptom	Fix
`cortex` not found	Add `~/.local/bin` to PATH; `chmod +x ~/.local/bin/cortex`
Install script fails	Re-run with `curl -q...` (ignores ~/.curlrc)
Connection error	Check `connections.toml`; `cortex -c <name>`
"Daily credit usage limit reached"	Admin raises/UNSETs parameter
Snowsight CoCo missing	Need `COPILOT_USER` + Cortex role; limit not 0; Copilot not opted out; trial needs card
Permission error (Snowsight)	Default role lacks privilege - switch/alter default role
Model unavailable	ACCOUNTADMIN: enable cross-region inference
Skill not triggering	Specific wording; name it; `/skill list`; exact `SKILL.md` filename
Subagent has no tools	Lowercase tool names
MCP won't connect	`/mcp`; `cortex mcp start`; check env vars; `~/.snowflake/cortex/logs/`
MCP OAuth fails	`cortex mcp remove` + re-add; free port 8585
ACP agent exits/garbled	Run `cortex acp serve -c X` in terminal; remove stdout writes from shell init
Automation "succeeded" but nothing happened	Read thread transcript; hooks can fail silently
Managed-settings banner "restricted mode"	Invalid JSON - run `cortex managed-settings`
Desktop sign-in browser won't open	Use "Browser didn't open?" link

21 · CAVEATS TO REMEMBER

- Vendor claims (72.1% vs 65.1% ADE-Bench; 7,100 accounts) are **unreplicated** - pilot on your own repos.
- Context stops at the Snowflake boundary (no native cross-system lineage/glossary).
- Preview: MCP, ACP, plugins, Agent SDK, automations. Skill Catalog rolling out.
- HIPAA / BAA / data-residency for CoCo: not covered by sources read - confirm with Snowflake & Compliance.
- Conversation files in `~/.snowflake/cortex/conversations/` may contain data - protect or use `--private`.

22 · SAFE-START WORKFLOW (10 steps)

1. Confirm role + connection: ask "What databases do I have access to?"
2. Use a role-per-environment connection; read-only role for prod.
3. Start restricted: `/guardrails` or `--with-restricted-session-scope=...`.
4. Check `AGENTS.md` / project conventions are loaded.
5. Switch to **Plan** (`Shift-Tab`/`Ctrl+P`) for anything multi-step or risky.
6. Name objects fully qualified: `DB.SCHEMA.TABLE`; `@`-attach context.
7. Review the plan, then approve; review diffs before accepting.
8. Ask for SQL used + assumptions + validation query; save to repo files.
9. Verify results independently (row counts, known report).
10. Check spend: usage views / daily limit; end with `/new` or `/clear`.

23 · WHERE TO LOOK

Need	Doc page (docs.snowflake.com/user-guide/cortex-code/...)
Flags/slash cmds	cli-reference
Skills/hooks/subagents	en/.../extensibility
MCP / ACP / plugins	cortex-code-mcp · cortex-code-acp · cortex-code-plugins
Automations	cortex-code-automations
Cloud sandbox	cloud-sandbox
Security / policy	security · managed-settings · mdm-policy
Cost & limits	cost-controls · credit-usage-limit · observability
Release notes	changelog (CLI) · cortex-code-desktop/release-notes
Desktop features	cortex-code-desktop/<feature>

Docs hub: docs.snowflake.com/en/user-guide/cortex-code/cortex-code · CLI ref: .../cli-reference · Changelog: .../changelog